Wanja Anastasia Tolksdorf Winter semester 2021/2022

Governance of Human-Carnivore Relations in Alpine Pasture Landscapes on the Example of the Return of Wolves in the French Alps

–

Towards a Resilient Coexistence

1. The problem of wolf-human conflicts

Since 1992 the grey wolf has been recolonizing the Western Central Alps (including French, Italian and Swiss territory) which contributes to the recovery of this endangered ecosystem keystone species (Sedy, 2018, pp. 2–3). But what seems a success story entails ongoing conflicts between the growing wolf population and pastoral farming which secures traditional livelihoods and preserves alpine grasslands (Sedy, 2018, p. 10). While protecting wolves from human threats like habitat loss and persecution, local communities must be protected from wolves predating on the farmers' livestock, competing with hunters and scaring residents (Sedy, 2018, pp. 2–3), to ensure a "resilient coexistence" (Pettersson et al., 2021, p. 2) in which humans and wolves can thrive. Another layer of the problem are the value conflicts between NGOs and the general population in favour of strict wolf protection and the affected rural communities calling for more extensive defence measures against wolves (Bertrand, 2021).

2. Existing governance attempts and their effectiveness

To alleviate the described problem, governance attempts on multiple levels already exist. On the *international level* the Convention on Biological Diversity, the Convention on International Trade in Endangered Species and the Bonn Convention on migratory species apply (Sedy, 2018, pp. 7–8).

Influential instruments at *European level* are the EU Habitats Directive and the Bern Convention. They define the legal status of the wolf as strictly protected with derogations still possible (Sedy, 2018, pp. 4–7). The Habitats Directive furthermore requires the establishment of Natura 2000 sites and reaching the favourable conservation status (Schnidrig et al., 2016, p. 13; Sedy, 2018, p. 4). The European Commission currently finances the project LIFE WolfAlps EU consisting of an international team of scientists, administrators and communicators that aims at coordinated actions to improve wolf-human coexistence at the alpine population level through monitoring, damage prevention, stakeholder engagement and more. (*LIFE WolfAlps EU*, 2020).

At the *level of the Alps* there is the Alpine Convention between the eight countries of the Alpine Arc which aims at sustainable development. It is implemented by a permanent committee and various expert working groups including the Large Carnivores, Wild Ungulates and Society Working Group which shall facilitate international cooperation to reach the favourable conservation status of large carnivores (Schnidrig et al., 2016, pp. 10–14).

Regarding *national level* approaches, the wolves in the French Alps are protected by French law, but culling of a limited number of livestock depredating individuals is allowed (Sedy, 2018, p. 4). There is a French national action plan for wolves by the environmental and the agricultural ministry about livestock protection, compensations, culling and monitoring (Sedy, 2018, p. 4).

The national wolf plan is implemented on the *departmental level* (Schnidrig et al., 2016, p. 18). Additionally, at (sub-)national levels there are different carnivore management boards and working groups of representatives of NGOs, the agricultural sector, public authorities and scientists for information exchange and consultation (e.g. Michaud, 2021; Sedy, 2018, pp. 8–9).

To summarize, the supranational governance focuses on wolf protection by legal means and facilitates transboundary collaboration. The favourable legislation is effective in protecting wolves and enabled, along with the recovery of forests and wild ungulates, their growth and recolonization of the French Alps (Schnidrig et al., 2016, p. 22; Sedy, 2018, p. 2). National and sub-national governance aims at resolving human-wolf conflicts and reach resilient coexistence. This has been less successful so far. Farmers are frustrated about increasing livestock predation and a rift between them and public authorities is crystallizing (Bertrand, 2021). Spokespersons of NGOs on the other hand point out the lack of willingness of farmers to implement protection measures (Bertrand, 2021).

3. Characterization of the problem in terms of governance concepts and their recommendations for alleviating the problem

Looking at existing approaches to govern the human-wolf relations it becomes evident that *multi-level governance* is employed. Multi-level governance structures can balance the benefits and drawbacks of centralization by assigning different responsibilities to different levels (Hooghe & Marks, 2003, p. 235). For example, wolf protection at European level has proven to be effective in ensuring the recovery of this border-crossing species. EU and national level governance also enable more efficient redistribution, for example in the form of compensation payments. The choice of specific measures for coexistence and their implementation, however, is better done on a more local level to adjust for ecological and socio-economic heterogeneity (Hooghe & Marks, 2003, p. 235; Sedy, 2018, p. 9). Concerning types of multi-level governance, the current wolf governance builds on an established Type I governance system (e.g. EU, national ministries, departments) with general-purpose jurisdictions and a nested design (Hooghe & Marks, 2003, p. 236). On top of that, it employs various Type II governance approaches (e.g. Bern Convention, Alpine Convention) that are more task-specific and intersecting (Hooghe & Marks, 2003, p. 236). There is a call for more transboundary approaches governing the whole wolf population, that means Type II governance (Sedy, 2018, p. 10).

High conflict levels and low acceptance of wolves by local residents can be viewed through the lens of *participatory governance*. Stakeholder involvement is so far restricted to the implementation of measures, education or consultation, but their involvement in decision-making seems rather low. It is not surprising that these top-down decisions are not well accepted by stakeholders. A participatory decision-making process could increase the legitimacy and the environmental outcome of the process. That is because a participatory process with stakeholder involvement, procedural fairness, involvement of addressees and collaborative decision-making is assumed to lead to higher acceptance of the output by stakeholders and better implementation and compliance with the output (Newig et al., 2018, p. 274). For example, the acceptance of wolf population growth and willingness to implement protection measures could be increased. However, the existing multi-level wolf governance system and policies limit the scope of decisions participants can negotiate on. Perhaps mandated participatory planning could give participatory governance a more central role in the already existing multilayer implementation approach (Newig & Koontz, 2014, pp. 259–260).

Participation and communication is also recommended from a *risk governance* view, especially in the case of highly ambiguous risks (Renn et al., 2011, p. 242). The wolf-human conflicts can be categorized as a highly ambiguous risk as different stakeholders have different perceptions about acceptable or tolerable levels of risk of wolf extinction as well as risk of predation (Renn et al., 2011, p. 235). Therefore, additionally to the current technical risk assessment based on monitoring population status and cases of predation (Sedy, 2018, p. 3), a concern assessment and a risk evaluation should be conducted to account for the normative dimension of the problem: the different values and worldviews of the stakeholders (Renn et al., 2011, pp. 239–240).

4. Conclusion

In conclusion, the improvement of the governance of human-wolf relations in the French Alps could lead to substantial improvements in the outcome and eventually to a resilient coexistence. This, however, does not imply a risk-free state like zero risk of depredation, but rather a tolerable level of risk while having effective institutions in place to cope with emerging conflicts (Pettersson et al., 2021, pp. 2–3). Given the high media attention, achieving human-wolf coexistence could function as a larger symbol for the compatibility of humans and nature.

References

Bertrand, S. (2021). TOUT COMPRENDRE – Attaques de loup, éleveurs à bout: pourquoi la situation se tend dans les Alpes du Sud. *BFM Dici*. https://www.bfmtv.com/bfm-dici/tout-comprendre-attaques-de-loup-eleveurs-a-bout-pourquoi-la-situation-se-tend-dans-les-alpes-du-sud_AN-202110210360.html

Hooghe, L., & Marks, G. (2003). Unraveling the Central State, but How? Types of Multi-level Governance. *American Political Science Review*, *97*(02), 233–243. https://doi.org/10.1017/S0003055403000649

LIFE WolfAlps EU: Coordinated actions to improve wolf-human coexistence at the alpine population level. (2020). https://www.lifewolfalps.eu/en/

Michaud, T. (2021). *Arrêté n° DT-21-0045: Portant composition, organisation et fonctionnement du comité départemental de suivi du loup dans le département de la Loire.* Direction Départementale des Territoires. http://www.loire.gouv.fr/IMG/pdf/ap_composition_comite_suivi_loup_loire.pdf

Newig, J., Challies, E., Jager, N. W., Kochskaemper, E., & Adzersen, A. (2018). The Environmental Performance of Participatory and Collaborative Governance: A Framework of Causal Mechanisms. *Policy Studies Journal: The Journal of the Policy Studies Organization*, *46*(2), 269–297. https://doi.org/10.1111/psj.12209

Newig, J., & Koontz, T. M. (2014). Multi-level governance, policy implementation and participation: the EU's mandated participatory planning approach to implementing environmental policy. *Journal of European Public Policy*, *21*(2), 248–267. https://doi.org/10.1080/13501763.2013.834070

Pettersson, H. L., Quinn, C. H., Holmes, G., Sait, S. M., & López-Bao, J. V. (2021). Welcoming Wolves? Governing the Return of Large Carnivores in Traditional Pastoral Landscapes. *Frontiers in Conservation Science*, *2*, Article 710218. https://doi.org/10.3389/fcosc.2021.710218

Renn, O., Klinke, A., & van Asselt, M. (2011). Coping with complexity, uncertainty and ambiguity in risk governance: A synthesis. *Ambio*, *40*(2), 231–246. https://doi.org/10.1007/s13280-010-0134-0

Schnidrig, R., Nienhuis, C., Imhof, R., Bürki, R., & Breitenmoser, U. (2016). *Wolf in the Alps: Recommendations for an internationally coordinated management.* RowAlps Report. KORA and BAFU, Switzerland. https://www.pyrenees-pireneus.com/Faune-Pyrenees/Loups/Europe/Suisse/2016-03-00-Wolf-Alps-Recommendations-internationally-coordinated-management-Loup-Alpes-Recommandations-gestion-coordonnee-niveau-international.pdf

Sedy, K. (2018). *Overcoming barriers - management of large carnivores in the Alps.* European Parliament. https://doi.org/10.2861/904779